



THE FIRELESS LOCOMOTIVE



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A new era arrives. A spotless Diesel is pushed into the NCR yard to replace the Company's venerable steam-storage locomotives.



On one of its last working days, the Rubicon, now on display at Carillon Park, pushed its successor to the factory engine house.



With several NCR employees aboard, the replacement engine makes its first run across Dayton's South Main Street.



The old yields to the new. The Rubicon turns switching responsibilities over to the modern Diesel. The transfer is symbolized by an employee handclasp.

“The Three little Engines”



The Rubicon, as yet unnamed, arrived at NCR in 1909 on a flat car, just as its successor was welcomed more than half a century later.

Most things yield to progress—and that, of course, is how museums are made. The Corliss engine now on display at Carillon Park labored mightily during the formative years of American industry, yet more economical electric power systems eventually sent that steam giant into retirement at the Park. The Conestoga wagon and the Concord coach, the Grasshopper locomotive and the high-wheeled Cadillac—all served their purpose. And

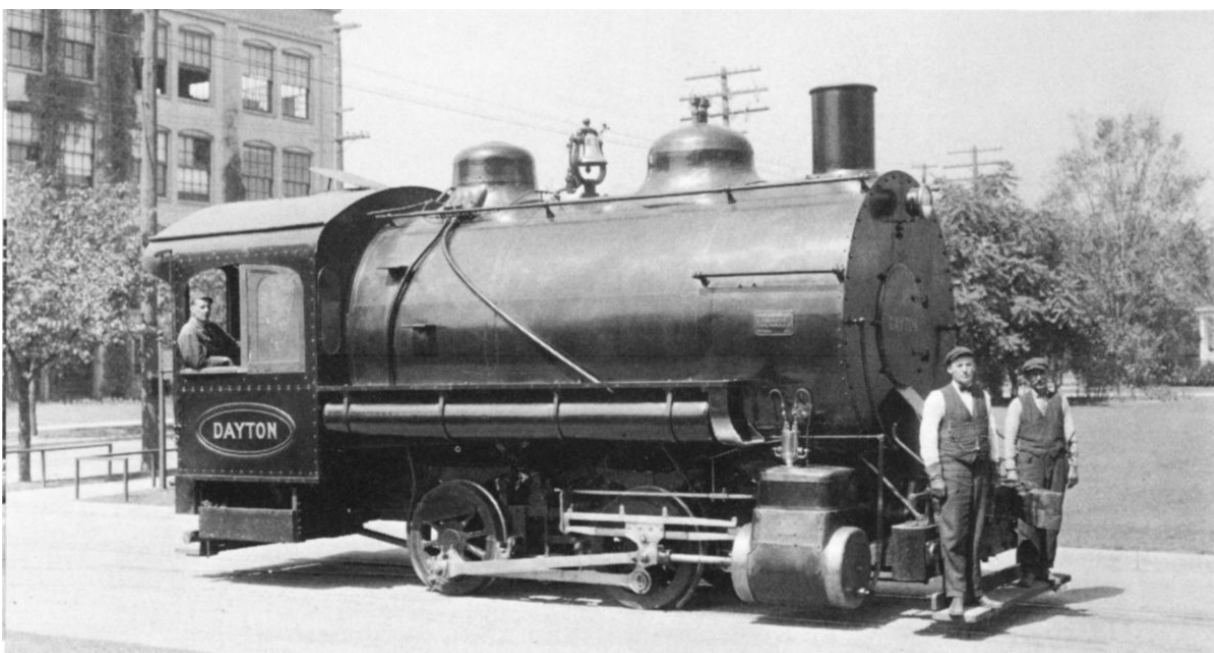
then, with sentimental if somewhat whimsical ceremony, they were consigned to their final resting place.

In the summer of 1962, Carillon Park made room for what is probably its most unusual example of antique “rolling stock.”

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“The Three Little Engines” are pictured soon after their purchase by NCR. The Carillon Park relic, the Rubicon, is pushing the first electric express car used by The Ohio Electric Company. The exhaust stacks of all three fireless engines originally extended up the front of the storage tanks, fully exposed, but the tanks were later given false fronts so the Rubicon, The Dayton and the South Park would resemble conventional railroad locomotives.



The Dayton



The South Park

The newcomer is the “Rubicon,” one of three fireless locomotives 3 which were purchased by The National Cash Register Company in the early years of this enterprising century. The Rubicon, however, did not

come to the Park from a rusty and forgotten limbo—as, for instance, the Grasshopper locomotive did. The engine is an antique, to be sure, yet its boiler and baffles had scarcely had time to dry when it was refurbished for a place of honor in the museum. Its successor, a Diesel-powered switch engine, had been delivered only a short time before the Rubicon was relieved of service and was ready to be converted into a public curiosity. In fact, the Rubicon—injury added to insult!—on one of its final trips puffed across the NCR yard and pushed its bright blue-and-yellow replacement from the flat car on which it arrived to the roundhouse.

“The Three Little Engines,” so long familiar to Daytonians, were among the first fireless (or steam storage) locomotives in America. The Dayton was built in 1913, the South Park in 1910 and the Rubicon in 1909—all by the Lima Locomotive Works of Lima, Ohio, on a basic design developed and popularized in Germany. NCR’s founder, John H. Patterson, had in fact seen such an engine during his travels in Europe, and decided it was just what he wanted for Dayton, Ohio.

Mr. Patterson was one of the first American industrialists to be concerned with “factory environment.” He believed that a factory would be an esthetic asset to the community, as well as a happy place for workers, if it were kept clean and attractively landscaped. NCR resembled—then, as it does today—a series of office buildings rather than a huge industrial complex.

The neighborhood adjacent to the factory also concerned John H. Patterson; he offered annual prizes to householders who kept the best yards and flower beds. An early NCR machinist who looked up from his lathe and glanced out the spacious window—by way of resting his eyes—saw lovely suburban gardens instead of the grimy clutter that bordered most factories of the era. 4

Keeping the sandstone buildings clean, and the geraniums healthy, would be easier, Mr. Patterson concluded, if it weren’t for the sooty smoke belched up by switch engines.

The steam-storage locomotive proved to be the answer. It could chuff about for hours, emitting nothing more than a few puffs of steam.

A Steam Tank on Wheels

The Rubicon is actually little more than a 7- by 16-foot steam tank, built somewhat like a thermos bottle and fitted out with cylinders and wheels. The tank was two-thirds filled with water and then charged from a 150-pound steam line from the NCR powerhouse.

As the Rubicon's engineer opened the throttle, steam passed through a reducing valve and reached the cylinders at a pressure of 60 pounds per square inch. The steam charge, at 370 degrees Fahrenheit, gradually converted some of the water to steam, which—although at a lesser pressure—gave the locomotive additional operating time.

The Rubicon ran three or four hours on a charge, depending on the work load. Normally, three or four daily trips were made to the roundhouse, to exchange a “tired” engine for a freshly-charged one. Eighteen-inch pistons enabled the engines to move their own weight with just a few pounds of steam, so it was rare for them to be stranded away from their “lifeline.” The storage tank was insulated with a two-inch layer of magnesia. It was fitted with baffles to keep the water from sloshing back and forth as the engine started and stopped.

Tanks under the locomotive running boards furnished compressed air for ringing the bell and sanding the rails in icy weather. They were replenished at the roundhouse, for the engine carried no air compressor. Neither was there a generator; the storage battery which operated the headlights had to be recharged regularly.

The engine's brakes were strictly mechanical—operated by tightening the large wheel in the cab.



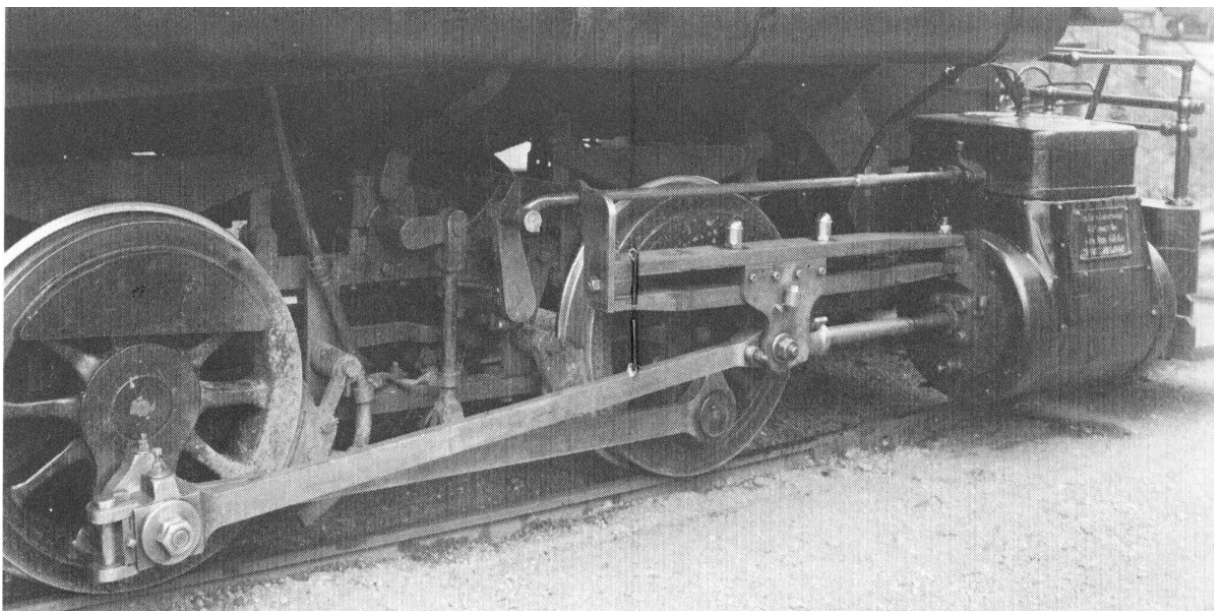
The Carillon Park museum piece was photographed at the NCR South Main Street factory crossing in the early years of the century. Seen at left is the NCR

office building, prior to construction of the NCR auditorium.

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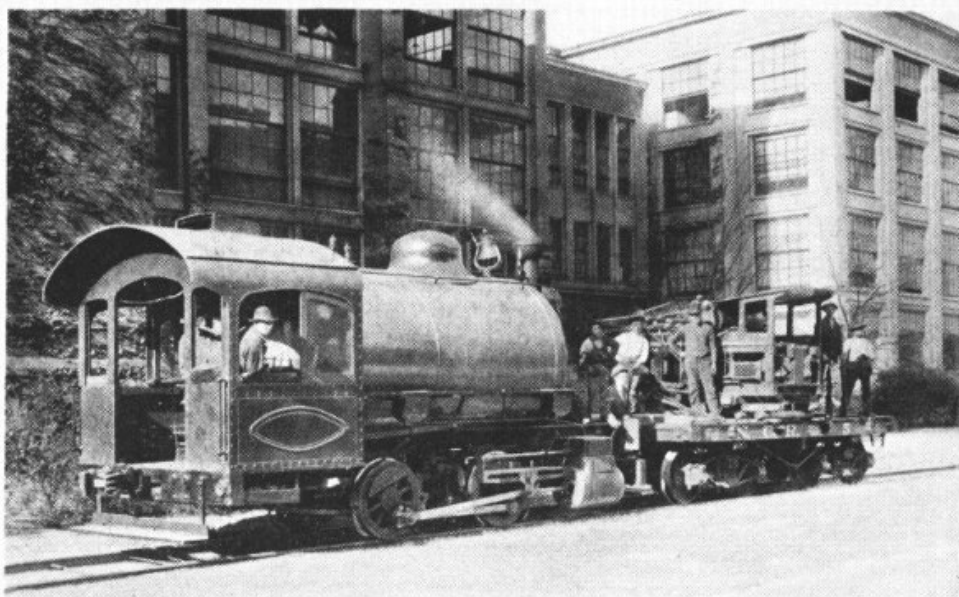
These pictures record an event which was perhaps inevitable with the increase of auto-age traffic; in 1915, a touring car slid into the Rubicon's side at the Main Street crossing.



The engine clearly won the contest!



It suffered only a slightly bent driving rod, visible in the photograph.



FIRST STEAM-STORAGE LOCOMOTIVE USED IN AMERICA, BUILT BY THE LIMA LOCOMOTIVE AND MACHINE CO., LIMA, OHIO, WORKING AT PLANT OF NATIONAL CASH REGISTER CO., DAYTON, OHIO

LIMA STEAM STORAGE LOCOMOTIVES

LIMA STEAM STORAGE LOCOMOTIVES consist essentially of a large tank, large cylinders, the other machinery being similar to that of regular locomotives.

The tank is filled about half full of water, and is then connected with a stationary boiler until the pressure equalizes.

When this occurs, considerable steam will have been condensed, but the water will have been raised to nearly the pressure and temperature of the steam in the boiler. As steam is used, the pressure falls, but with the decrease part of the water becomes steam. The tank is charged to full boiler pressure, 125 to 200 pounds, as the case may be, and the pressure reduced to 60 pounds by a reducing valve.

The cylinder diameter is increased so the tractive power, to the limit of adhesion, can be utilized at 60 pounds pressure in the cylinders. Due to these large cylinders, the locomotive can move itself with only 3 or 4 pounds pressure.

Under ordinary circumstances, it will not have to be charged any oftener than the regular type of locomotive takes water. Varying with the amount of work desired, it will run from two to ten hours with one charge—two charges per day is a good average.

This type possesses many advantages for work in industrial plants, powder mills, lumber yards, cotton mills, wharves, etc. Among its advantages are:

1. **ABSOLUTE SAFETY** in inflammable localities, and from boiler explosion, as pressure decreases constantly.
2. **SIMPLICITY**.—Nothing to watch but signals and gauge.
3. **ECONOMY**.—In first cost; in maintenance; in operation.

Just the locomotive for use around your plant. Absolutely no danger from fire; can be operated around factories manufacturing the most inflammable materials with perfect safety. We build all practical sizes. Write for further particulars.

The Lima Locomotive Works, manufacturer of the Rubicon, featured a photograph of the busily puffing engine in one of its catalogues. The page is

reproduced here.

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Lima steam storage locomotives consist essentially of a large tank, large cylinders, the other machinery being similar to that of regular locomotives.

The tank is filled about half full of water, and is then connected with a stationary boiler until the pressure equalizes.

When this occurs, considerable steam will have been condensed, but the water will have been raised to nearly the pressure and temperature of the steam in the boiler. As steam is used, the pressure falls, but with the decrease part of the water becomes steam. The tank is charged to full boiler pressure, 250 to 200 pounds as the case may be, and the pressure reduced to 60 pounds by a reducing valve.

The cylinder diameter is increased so the tractive power, to the limit of adhesion can be utilized at 60 pounds pressure in the cylinders. Due to these large cylinders, the locomotive can move itself with only 4 pounds pressure.

Under ordinary circumstances, it will not have to be charged any oftener than the regular type of locomotive takes water. Varying with the amount of work desired, it will run from two to ten hours with one charge. Two charges per day is a good average.

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1. ABSOLUTE SAFETY in inflammable localities, and from boiler explosion, as pressure decreases constantly.
2. SIMPLICITY. Nothing to watch but signals and gauge.

3. ECONOMY. In first cost; in maintenance; in operation.

Just the locomotive for use around your plant. Absolutely no danger from fire; can be operated around factories manufacturing the most inflammable materials with perfect safety. We build all practical sizes. Write for further particulars.

The fireless “locos” were undeniably safe, in that steam pressure was always on the decrease and never ran wild. On the other hand, the brakes were so primitive that fast emergency stops were impossible with a string of heavy coal cars. Visibility from the cab was limited, too—an added hazard with increased auto and pedestrian traffic in the NCR factory area in recent years. 9

But the retirement of the Rubicon, the Dayton, and the South Park was irrevocably decided by the mounting cost of keeping them in repair. Replacement wheels had to be specially cast, and many other parts had to be fashioned from scratch. Toward the end of the engines’ service, two machinists were devoting full time to pampering their aches and agues, and the maintenance bill was coming to more than \$16,000 annually. Railroad buffs will miss them, but they had to go!



The NCR engine house had just been completed when employees clustered around the Dayton for this photograph.



One of the first engine crews poses by the Rubicon.

The Diesel-electric locomotive which replaced them is a 50-ton eight-wheel unit manufactured by the General Electric Company at Erie, Pa. It is powered by two 150-horsepower engines, each driving a D.C. generator. In turn, each of the generators powers a D.C. motor geared to an axle. A powered axle drives two wheels directly, and by side arms two other wheels are also driven. Thus, the locomotive is powered by two independent units. The power units can be used separately or simultaneously as the number of cars demands, giving traction to either four or eight wheels.

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NCR helped Dayton meet the 1913 flood emergency. The Company's powerhouse—equipped with the giant Corliss engine which is seen today at Carillon Park—supplied the stricken city with electricity.

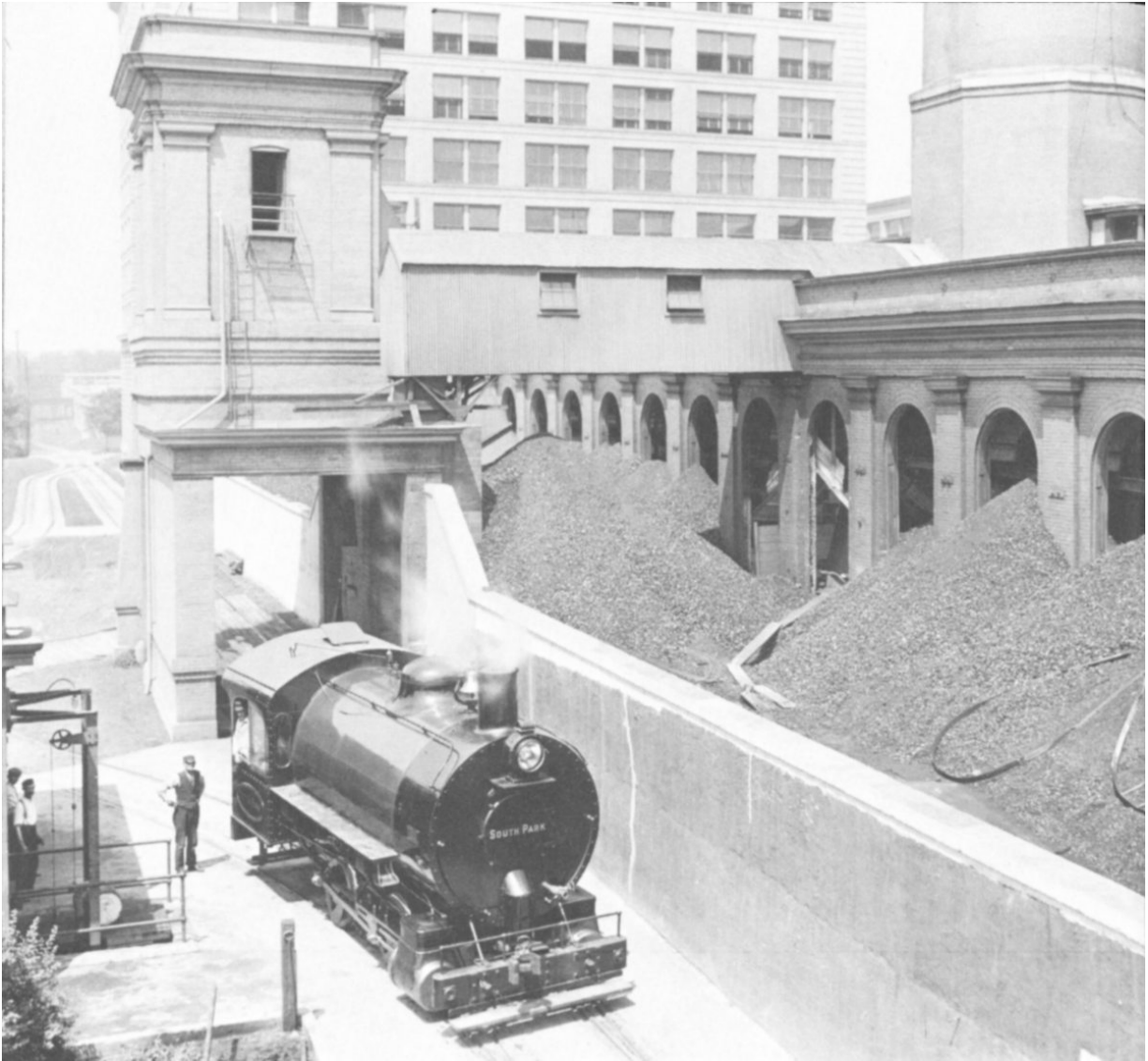


Moreover, the hard-working Rubicon was sent north on the streetcar tracks to help haul away flood debris.

With the Diesel-electric, no time is lost re-charging the tank. The engineer's cab is comfortably heated in winter. Automatic couplers, front and rear, increase employee safety. Powerful air brakes control the engine itself and air hoses can also be coupled to towed cars. Operating cost is only a fraction of that required for the old "chuffers."

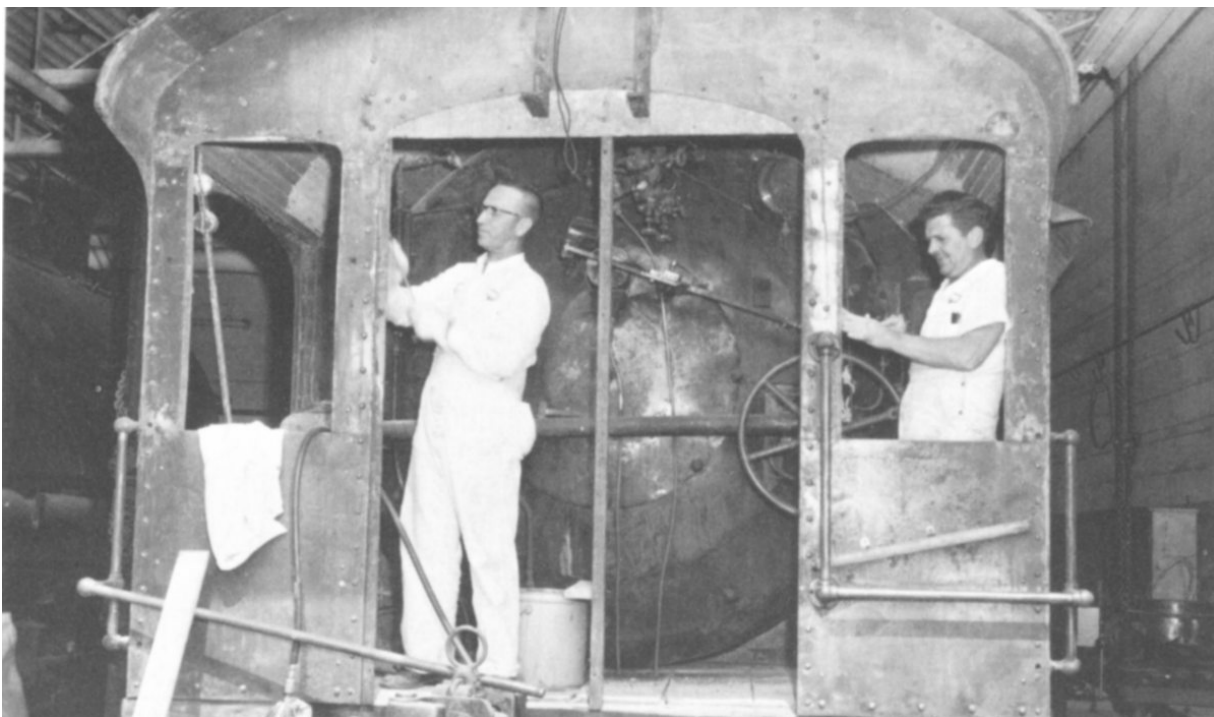
Fireless Engine Slighted in Rail Lore

The steam storage locomotive does not figure prominently in the colorful literature of railroading. Confined to the modest task of shunting miscellaneous cars about remote factory yards, there was no Casey Jones to give it romance. Nor was there a lusty fireman or a wandering hobo to immortalize it in song. It is known that an obscure “Toonerville” type of road near New Orleans employed a fireless engine as early as 1835—12 recharging at each end of the track. Also, a number of fireless engines were used around paper mills and munitions factories, where sparks from conventional engines could have led to fire and cataclysm.



The South Park, sister engine of the Rubicon, edges up to the NCR powerhouse. The photograph is undated, but the gleam of the engine indicates that it hadn't seen too many years of service.

The Rubicon is one of the last of its breed—perhaps even the last of its particular design. But sporting new black paint and fresh gold lettering, it has found a measure of immortality—albeit without balladry—at Carillon Park, among other relics of America's industrial past.



After the Rubicon's retirement, workmen began refurbishing it for the Carillon Park historical collection. Here they are seen removing half a century's accumulation of paint.



The Rubicon is shown as it traveled the last few feet of its long career. It was moved into its permanent Carillon Park home on July 16, 1962.



On August 20, 1962, the Engine House at Carillon Park was opened with informal ribbon-cutting ceremonies. Shown above are several of the guests inspecting the refurbished locomotive.

CARILLON PARK
DAYTON, OHIO

One of a series of Carillon Park booklets.
Price ten cents.

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